Jodh Singh Nahal

778-538-2351 | nahal.jodhsingh@gmail.com | linkedin.com/in/jodhsinghnahal | github.com/jodhsinghnahal

Education

University of British Columbia (UBC) *Vancouver, BC*
Bachelor of Applied Science in Computer Engineering *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

Technical Skills

Robotics: ROS/ROS2, OpenCV, YOLOv8, PyTorch, MuJoCo, LiDAR, SocketCAN
Embedded: Zephyr RTOS, FreeRTOS, CAN-FD, SPI, I2C, UART, STM32, ESP32, TI C2000, Teensy
Programming: C/C++, Python, MATLAB & Simulink, ARM Assembly
Tools: Linux, Git, Docker, GDB, JTAG, Logic Analyzers, Oscilloscopes, KiCad

Experience

Hardware Verification Engineering Intern *Vancouver, BC*
AMD *Sept 2026 – Present*
- Verifying the Input–Output Memory Management Unit L1 cache behavior for the AMD IOHUB architecture using UVM constrained-random and directed testbenches.
- Writing SystemVerilog Assertions (SVA) to cross-check legal signal responses across interconnected IPs.

Software Test / Embedded Development Co-op *Burnaby, BC*
Xantrex *May 2024 – Aug 2025*
- Developed object-oriented, multi-threaded C++ firmware on Linux-based Raspberry Pi gateways, achieving a 7× increase in device capacity and 90% reduction in CPU usage over the prior Python implementation.
- Developed Hardware-in-the-Loop (HIL) CAN and UART simulators in Python/C++ to execute 100+ tests for embedded gateway systems, validating edge cases and stress-testing real-time firmware performance.
- Profiled the C++ firmware to eliminate performance bottlenecks.
- Wrote Python scripts to autogenerate scalable C++ object-oriented code from SQLite and Microsoft Access/Excel data, eliminating manual boilerplate.

Embedded Systems Developer *Vancouver, BC*
UBC Rover *Sept 2023 – July 2026*
- Led a team of 3 engineers to develop a distributed embedded control system integrating ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C with aggregated data streamed over UART.
- Developed Zephyr RTOS firmware on a Teensy 4.1 for stepper-based control of a 6-axis robotic arm.
- Led migration to a CAN-FD architecture using Moteus r4.11 controllers and BLDC motors for closed-loop torque control and higher bandwidth.
- Interfaced an Intel NUC (onboard PC) with SocketCAN and bridged data to ROS.
- Built a ROS–Arduino pipeline using rosserial, OpenCV, and Python nodes to modulate USB camera framerates from potentiometer readings.
- Developed a 16-channel telemetry system on a TI C2000 (LAUNCHXL-F28069M) using ePWM-triggered ADC interrupts, streaming GPIO/ADC data over SCI/UART.
- Debugged C2000 firmware over JTAG in CCSTUDIO.

Undergraduate Research Assistant *Vancouver, BC*
UBC ECE — Supervisor: Dr. Lutz Lampe *May 2026 – Sept 2026*
- Designed power-management PCB hardware in KiCad integrating AC–DC rectifiers and switching circuitry.
- Simulated and optimized piezoelectric energy harvesting circuits in LTspice and Simulink/Simscape.

Projects

Autonomous Racing and Traffic Sign Recognition (F1Tenth) | *Python, YOLOv8, ROS, OpenCV*
- Architected a multi-node ROS2 stack with LiDAR and/or camera gap/wall-following and real-time command arbitration for collision-free racing on a Jetson Nano car.
- Engineered a hierarchical safety system with AEB and adaptive speed control for high-speed obstacle avoidance.

Behaviour Cloning for Robotic Control | *PyTorch, MuJoCo, NumPy, dm_control*
- Designed a 2-layer DNN in PyTorch mapping robotic state observations to joint torque predictions for a simulated Reacher task.